

Final Project

Text-Tabular Learning

AT A GLANCE

 Deadline	Monday, October 26, 2026 (all deliverables)
 Submission	Individual work (no pairs), via the course Moodle
 Tracks	Choose one : Track 1 (Algorithm) or Track 2 (Benchmark)
 Key lecture	July 19, 2026: MulTaBench, presented by Alan Arazi
 Final report	Short paper in English, up to 4 pages, NeurIPS template
 AI tools	Permitted and encouraged; you are responsible for everything you submit

1 Overview & Background

In many real-world applications, data does not arrive in a single modality. Instead, it is often structured in tables (tabular data) where some columns contain standard numerical or categorical features (e.g., a patient's age or blood type), while other columns contain free-text descriptions, medical notes, or product reviews. This setting is known as **text-tabular learning** (a sub-field of Multimodal Tabular Learning).

Standard tabular algorithms, such as Gradient Boosted Decision Trees (GBDTs like XGBoost [9], CatBoost [7], and LightGBM [8]) or Tabular Foundation Models (TFMs like TabPFN [3, 4] and TabICL [10, 11]), are natively unimodal and cannot process raw text. To solve text-tabular problems, practitioners must either:

1. **Represent the text via static embeddings:** Use a pretrained text encoder (e.g., E5 [12] or RoBERTa [14]) to generate embeddings, project/compress them, and append them as numerical columns alongside the structured features.
2. **Build joint models:** Train a dedicated architecture that natively fuses text representations with tabular features. Some joint models additionally allow the text encoder to adapt to the prediction target via **Target-Aware Representations (TAR)**, as in [TabSTAR \[2\]](#).

On **July 19th**, we will have a dedicated lecture on text-tabular prediction problems. During this lecture, **Alan Arazi** will present the [MulTaBench \[1\]](#) paper, which introduces a curation pipeline and benchmark for evaluating multimodal tabular learning models.

3 Track 2: The Benchmark Track

3.1 Objective

In this track, your task is to identify, preprocess, and evaluate a new dataset to determine if it satisfies the curation criteria of a text-tabular learning task. You will analyze your dataset's properties by applying the curation pipeline developed in the [MulTaBench \[1\]](#) study.

3.2 MulTaBench Curation Pipeline

To verify if a text-tabular dataset carries the necessary signal, you must evaluate it across **5 diverse tabular learners**: TabM [6], CatBoost [7], LightGBM [8], TabPFN 2.0 [4], and TabPFN 2.5 [5]. To run this evaluation, use the official [MulTaBench](#) source code, which already implements all five learners and all four experimental conditions described below.

For each learner, evaluate the performance under **4 distinct conditions**:

- **Unimodal Structured**: Tabular features only (numerical and categorical).
- **Unimodal Unstructured**: Text features only.
- **Joint Frozen**: All features combined, using static, frozen text embeddings (e.g., E5-small-v2 [12]).
- **Joint Target-Aware (TAR)**: All features combined, where the text encoder has been fine-tuned (e.g., using LoRA [13]) on the target variable.

3.2.1 Curation Criteria

For a dataset to successfully pass the curation pipeline, it must satisfy both of the following conditions for **at least 3 out of the 5** tabular learners:

1. **Joint Signal**: The performance under the *Joint Frozen* condition must be strictly higher than both the *Unimodal Structured* and *Unimodal Unstructured* baselines. This ensures that both modalities contain complementary predictive signals and prevents the dataset from collapsing into a pure NLP task.
2. **Task-Awareness (TAR Gain)**: The performance under the *Joint TAR* condition must be strictly higher than the *Joint Frozen* condition. This isolates the empirical benefit of target-aware representation tuning.

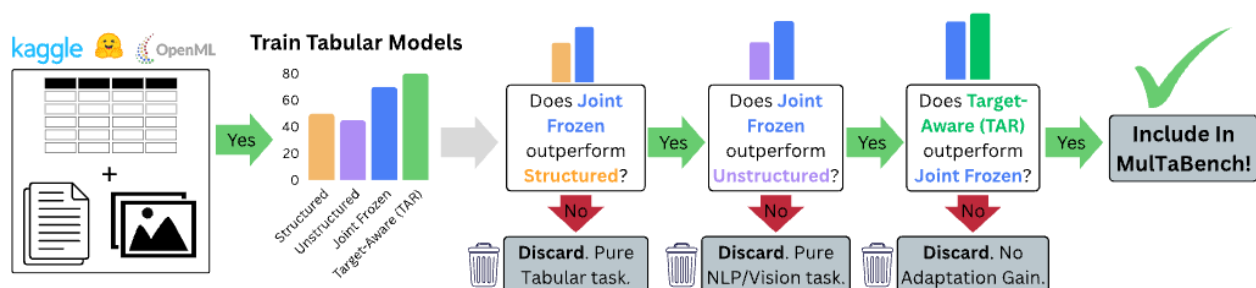


Figure 1: The MulTaBench Curation Pipeline. Datasets are included if joint prediction outperforms unimodal baselines and if Target-Aware Representations (TAR) improve on frozen, off-the-shelf embeddings.

DATASET EVALUATION & GRADING

The magnitude of the performance delta between the conditions will be the primary metric used to evaluate and grade the quality of your dataset. A larger, robust performance gap demonstrates a more successful and challenging curation process, which will reflect positively on your final project grade.

★ FUTURE COLLABORATION OPPORTUNITY

In the future, the course staff may collect the highest-quality datasets compiled by students in this track to build a larger, combined text-tabular benchmark. If a dataset you curate is included in this future benchmark, you will be invited to join as a co-author on the resulting academic publication.

3.3 Project Scope and Grading Tracks

To help guide your work, the final project can be completed at two different levels of scope:

3.3.1 Standard Project (Default Target)

The standard requirement is to identify, preprocess, and evaluate **one** new text-tabular dataset that meets the MulTaBench curation pipeline criteria. You must present evaluation results showing it satisfies the joint signal and task-awareness conditions.

- *Ideas for Standard Projects:*
 - Clean and restructure a raw public dataset (e.g., from Kaggle or OpenML) that contains text fields but is not currently part of MulTaBench.
 - Inject informative text features into a structured tabular dataset.
 - Incorporate structured features (e.g., metadata, user stats) into an NLP dataset.

3.3.2 Outstanding Project (Optional Extension)

For students looking for an extra challenge, you may propose an outstanding project. This requires developing a systematic method to automatically generate or mine new text-tabular datasets, and submitting at least **5 new datasets** generated or discovered via your method that successfully pass the MulTaBench curation criteria.

- *Ideas for Outstanding Projects:*
 - **Tabular-to-Text Enrichment:** Build a general pipeline that automatically enriches numerical/categorical tables with relevant text descriptions mined from the web.
 - **Generative Tabular Synthesis:** Use Large Language Models (LLMs) to synthesize realistic text-tabular datasets from scratch.
 - **Automated Dataset Mining:** Build a web crawler that automatically discovers raw text-tabular data sources, parses them, and checks if they pass the curation filters.
 - **Image-Text-Tabular Datasets:** Curate datasets that combine all three modalities (images, text, and structured features) and pass the curation criteria for *all* modalities, as in the